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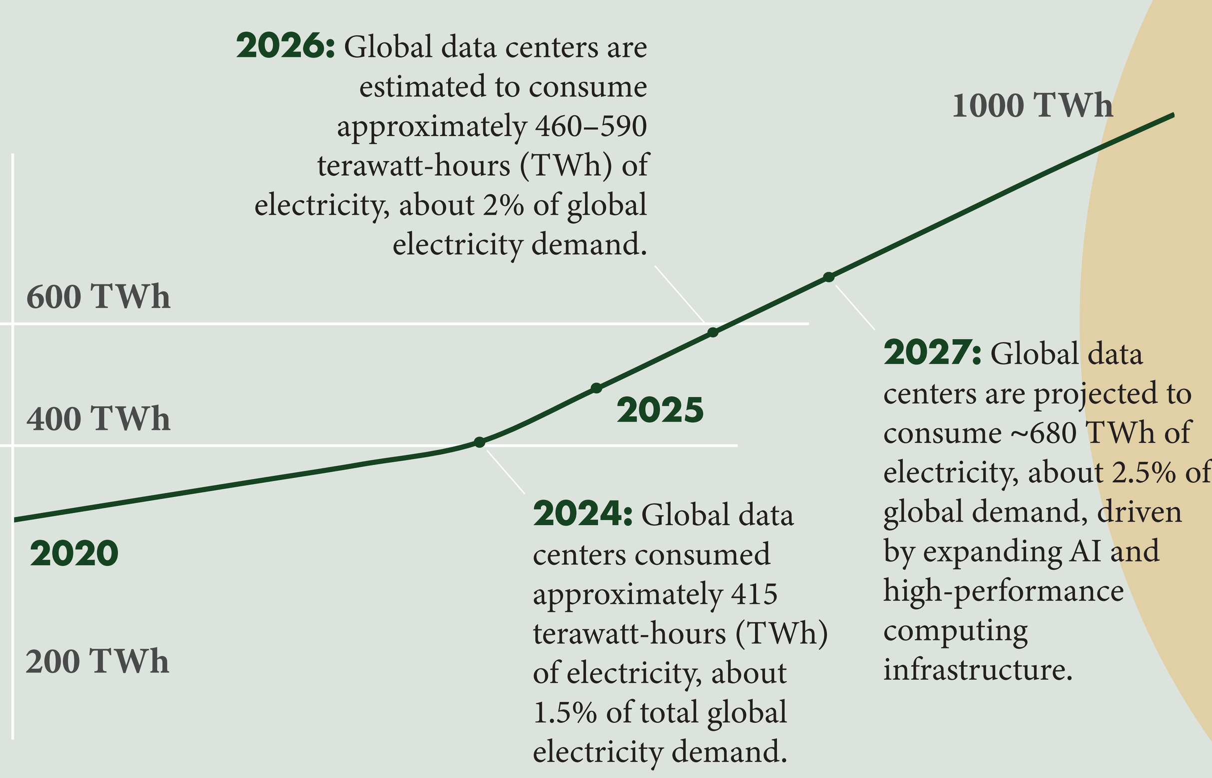
Less Power, Same Intelligence

Large Language Models (LLMs) and Their Growing Energy Demand

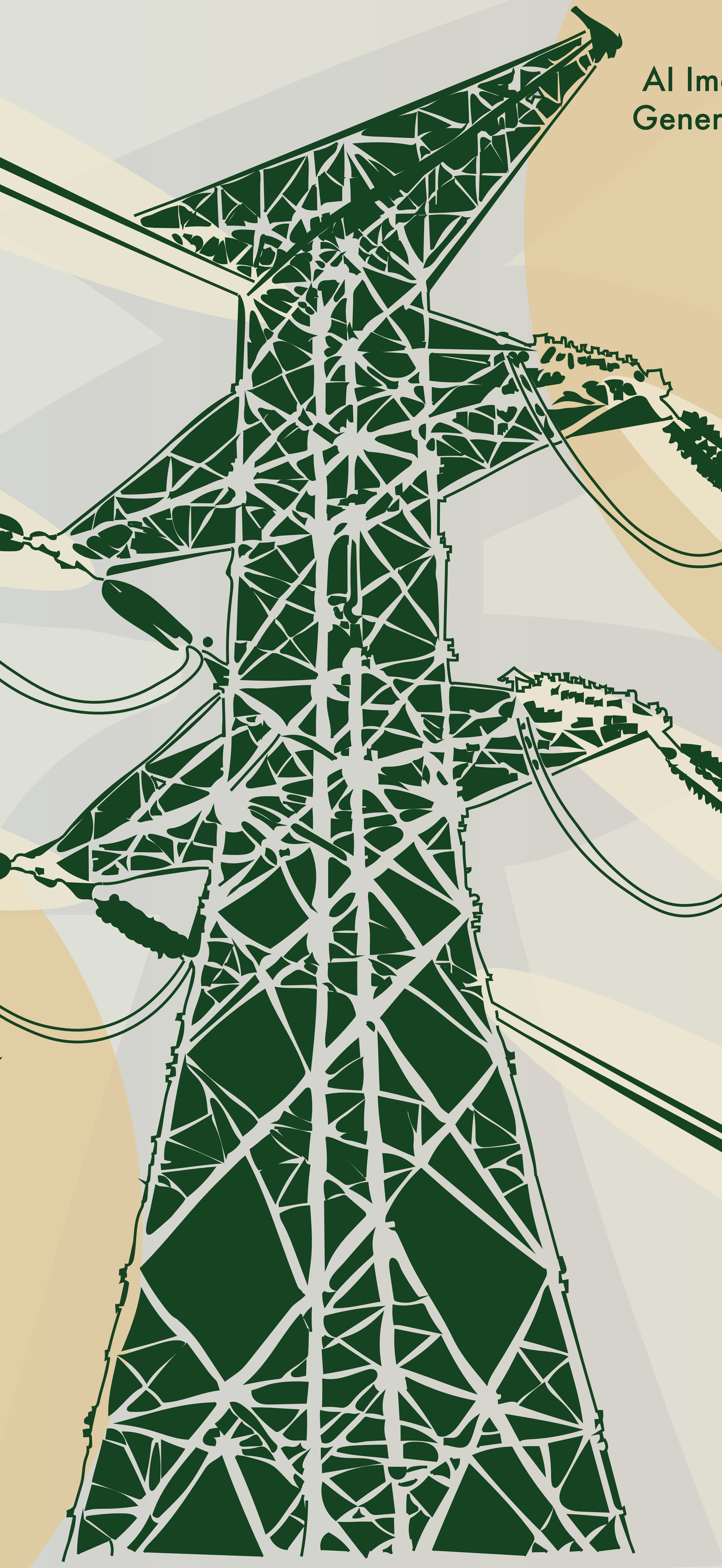
Large Language Models (LLMs) such as ChatGPT require enormous computing power to train and operate. These systems run on massive data centers filled with thousands of specialized graphics processing units (GPUs) that process calculations every second. Unlike a normal web search, LLMs generate responses token by token through continuous neural network computations, making AI significantly more energy-intensive than traditional software.

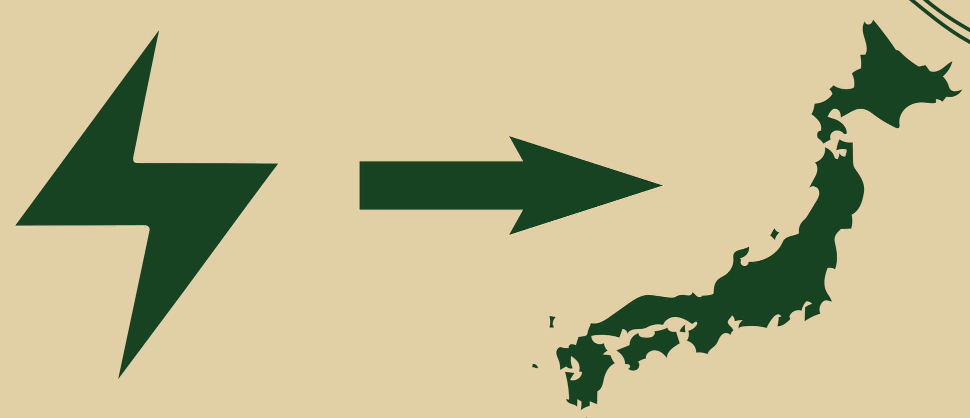
According to the International Energy Agency (IEA), electricity demand from data centers is expected to more than double by 2030, with AI identified as the most significant driver of this growth. In advanced economies, data centers are projected to account for nearly half of the increase in electricity demand between now and 2030. highlighting the growing influence of AI on global energy systems.

Growth in Data Center Power Consumption (TWh) During the AI Boom (2020–2027)

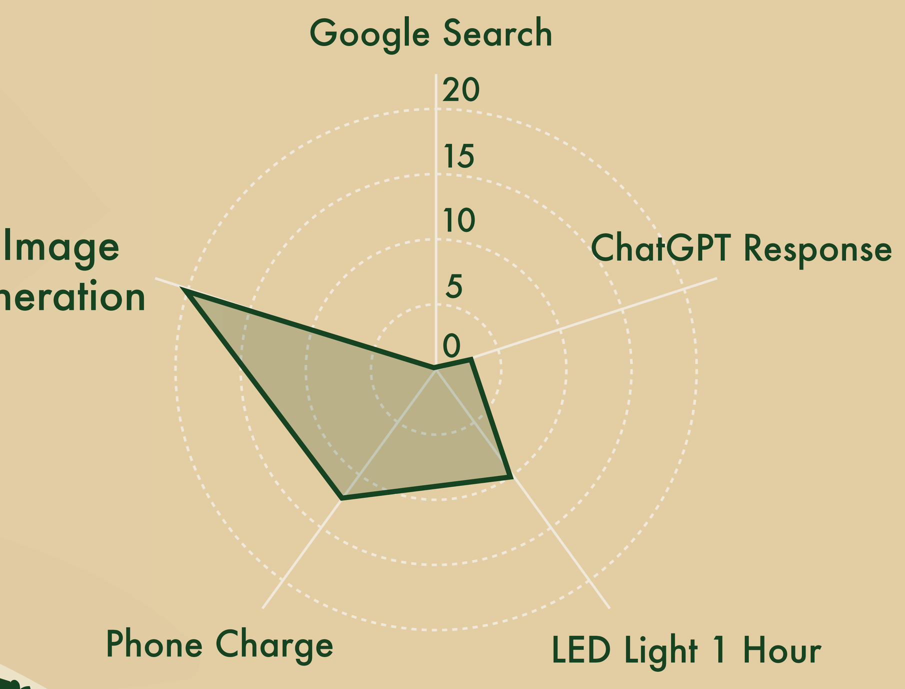


*Global data center electricity consumption is projected to **more than double** between **2020** and **2030**.*





*According to the International Energy Agency (IEA), electricity consumption from data centres, AI, and cryptocurrency is projected to reach approximately **945 TWh** by **2030** - about the same as **Japan's total annual** electricity consumption today.*



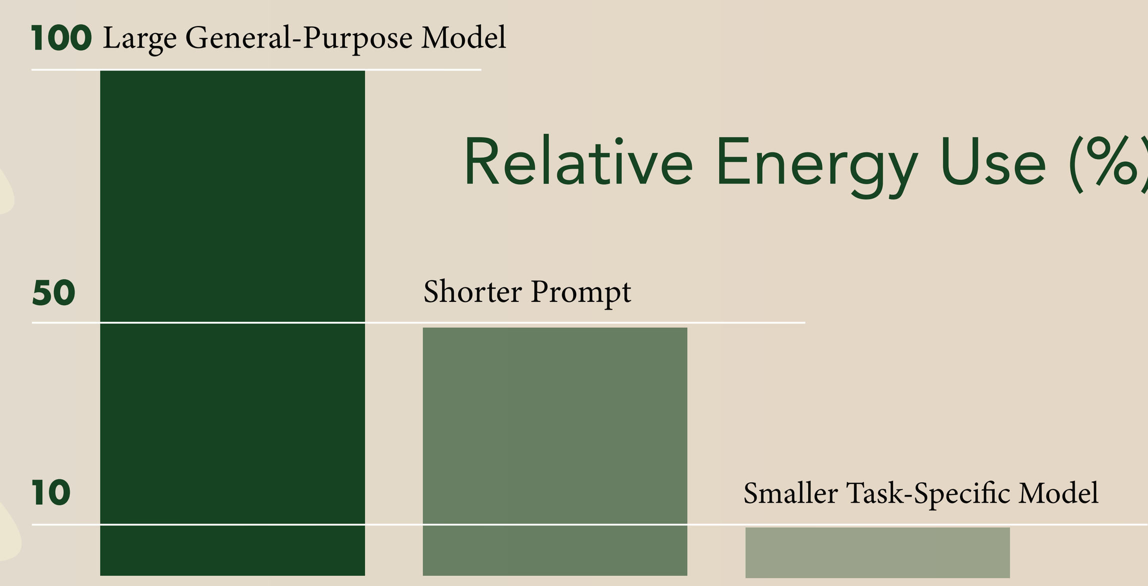
*Values are approximate and intended for comparison. Actual energy use varies by **hardware, model, query complexity, and data center efficiency.***

Sustainable AI Starts With You

AI energy use is strongly driven by how many tokens are processed per query, making efficiency one of the simplest ways to reduce its impact. A recent study conducted by Microsoft estimates that large language model inference consumes about 0.31–0.34 watt-hours per query under realistic conditions. Because energy use scales with token production, longer prompts and responses directly increase electricity consumption. The same study shows that "reasoning-style" or long-generation queries can consume approximately 13 times more energy than standard queries due to significantly higher token counts.

This means that small behavioral changes, such as writing concise prompts, limiting response length, and avoiding repeated regenerations can reduce energy demand per interaction. Organizations can further improve efficiency by deploying smaller, task-specific AI systems rather than relying on large general-purpose models for every request. For example, customer service chatbots can be designed to answer only predefined support questions, reducing unnecessary computation while still meeting user needs.

According to a joint UNESCO and University College London report, using smaller AI models tailored to specific tasks can reduce energy consumption by up to 90% without sacrificing performance. At scale, these efficiency gains matter because billions of AI queries translate even small per-request savings into substantial reductions in total data center electricity consumption.



*Task-specific AI models are designed for a limited set of functions and can consume up to **90% less** energy than large general-purpose AI models while delivering comparable performance.*

AI data centers require continuous power and cooling, placing additional strain on regional energy systems and increasing operational costs. In response, utilities and policymakers in many regions are planning upgrades to electricity generation, transmission, and distribution networks to accommodate growing demand. As Kaveh Madani, lead author of a United Nations University report, explains, "the public debate still often treats AI as software, but AI is also physical infrastructure: data centers, electricity generation, cooling systems, transmission networks, chips, minerals, land and water."

As discussions about AI's energy footprint continue, researchers have begun modeling how AI could shape future carbon emissions. According to Turkay et al. (2026), global AI-related emissions may follow three pathways: Baseline Growth, where rapid AI adoption drives increasing energy demand; an Efficiency Scenario, where advances in hardware and model design reduce energy use; and a Renewables Scenario, where data centers are powered by low-carbon electricity. The differences between these pathways demonstrate that AI's environmental impact is not predetermined.

Annual global AI-related carbon emissions in $MtCO_2$ (2020-2030) projected

*Projected AI-related carbon emissions vary often depending on how AI is developed and powered. Improvements could reduce emissions by **more than half** compared to a business-as-usual growth scenario, highlighting the importance of sustainable AI practices.*

The future of sustainable AI will be shaped by advances that cut energy use at every layer, from model design to data-center infrastructure. As providers expand access to renewable-powered data centers and organizations adopt smaller, task-specific models, the carbon intensity of AI systems is expected to decline even if usage grows, provided efficiency is prioritized throughout AI's development. In addition, system-level innovation will play an equally critical role in lowering AI's overall environmental footprint.

Sources: International Energy Agency (IEA), Turkay et al., 2026, United Nations, UNESCO & University College of London, Microsoft